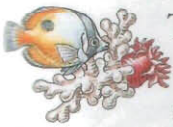


CORAL REEFS



TEEMING WITH WILDLIFE from shrimps to sharks, coral reefs are some of the most beautiful underwater structures. A coral reef takes thousands of years to form; it is composed of the living and dead skeletons of colonies of tiny animals called corals. Corals have flourished and built reefs in shallow, tropical seas for more than 440 million years. Corals are related to sea anemones and jellyfish, and belong to the group of animals called coelenterates. Australia's Great Barrier Reef is 2,000 km (1,240 miles) in length but is under serious threat from increased pollution.

Corals compete for light and food-bearing water currents.



Seahorse

Seahorses are pipe fish with upright S-shaped bodies and prehensile tails. They eat small crustaceans, such as shrimps, and suck them into their tubular mouths. Females lay eggs in the males' brood pouches; the young develop and later emerge.

Black tip reefshark

The largest inhabitants of the reef are the black tip reefshark and its distant relative, the giant manta ray. They patrol the seaward drop-off flank of reefs, on the lookout for potential fish prey that stray too far from the protection of the reef. By contrast, mantas are filter feeders attracted to the reef by the upwelling of plankton.

Fish

Coral reefs are home to fish of all shapes and sizes, from solitary giant groupers and wrasse, at up to 1.8 m (6 ft) long, which lurk in reef caves and recesses, to shoals of tiny damselfish, which graze the fronds of seaweed on the top of the reef. Some, such as the moray eel, are vicious predators; others such as parrot fish, feed on coral.



Mandarin fish

The most poisonous fish of the reef are the various scorpion fish, such as the mandarin. Their bodies are protected by bony plates and poison-tipped spines. Mandarin fish swim slowly. They live near the seabed, waiting to pounce on their prey.



FIND OUT MORE

Coral reefs

Coral reefs cover 619,000 sq km (239,015 sq miles) of the Earth's surface. Fringing reefs grow along coastlines, atolls are reefs that grow around extinct volcanoes, and cays are complete islands made of coral. Coral comes in all colours from red and yellow to blue and green, and grows in a variety of shapes and sizes including delicate fan corals, upright staghorn corals, and dome-shaped brain corals. Many animals hide in the holes and crevices within the reef.



Staghorn coral releasing eggs

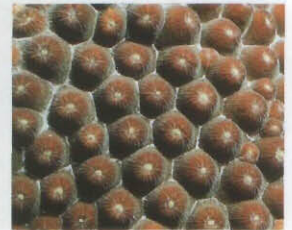
Reproduction

Corals reproduce asexually to form colonies of genetically identical polyps. They also reproduce sexually, releasing eggs and sperm into the water. The fertilized eggs turn into larvae that join the plankton, which is carried on the currents, ensuring wide distribution.

Symbiotic algae live within corals and give them their bright colours.



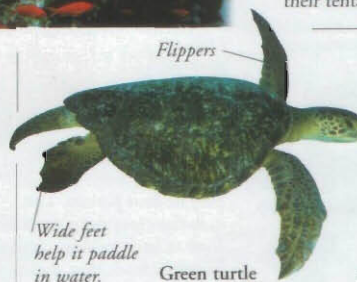
Coral polyps open at night



Coral polyps close in the day

Coral polyp

Coral reefs are made up of many individual animals called coral polyps. They normally live in large colonies, but a few species are solitary. Soft corals have a rigid inner layer for support; hard corals secrete a cup-shaped chalky skeleton from their jelly-like bodies. A circle of tentacles, armed with stinging cells, immobilizes prey and pushes it through the mouth into the stomach. Coral polyps close up during the day, but at night the reef comes to life when the corals extend their tentacles into the water currents ready to trap prey.



Green turtle

Reptiles

Sea turtles and sea snakes are reptilian inhabitants of reefs. Both need to surface to breathe air, but while turtles have to move onto land such as oceanic island reefs to breed, sea snakes can give birth to live young at sea. Some turtles prey on other animals, while others feed on grass. Sea snakes are good swimmers and are the most venomous snakes in the world. They prey on the abundant reef fish.

Invertebrates

Reefs provide a variety of habitats for invertebrates such as sea slugs, sea cucumbers, and sea urchins. The reef protects delicate filter feeders, such as sponges, from the impact of the waves. Some bivalve molluscs nestle in crevices; others such as mussels, anchor themselves to the coral with root-like hairs. Many crustaceans, such as shrimps, scavenge amongst the corals.



Common octopus

The reef is an ideal environment for the octopus to hide – its soft body slips easily into crevices. A stealthy hunter, it grabs prey with sucker-covered tentacles. When out in the open, its ability to change the texture and tone of its skin provides excellent camouflage.

Giant clam

This biggest living shellfish grows to 1 m (3 ft) long and, like corals, is inhabited by symbiotic algae. A clam opens its shell to feed on plankton and closes it if danger threatens. Giant clams remain stationary once they are adult.

Shell is made of two halves.



Siphon for waste

Algae

AUSTRALIA

FISH

OCEAN
WILDLIFE

OCTOPUSES
AND SQUIDS

SHARKS
AND RAYS

TURTLES AND
TORTOISES